

4(a)	less / no, oxygen for (aerobic) respiration ; cells / tissues / organs, fail to function / die ;	2
4(b)(i)	<i>any two from:</i> <i>if child has, sickle cell anaemia / SCT</i> 1 informed decision can be made about continuing with the pregnancy / AW ; 2 treatment can begin quickly (after birth) ; 3 parents can prepare, mentally / emotionally / for the cost of treatment ; 4 (this) child can make an informed decision about having children when older ; <i>if child does not have, sickle cell anaemia / SCT</i> 5 removes worry / AW ;	2
4(b)(ii)	<i>any three from:</i> <i>why</i> 1 (primers) anneal / bind, to DNA ; A hybridises 2 so that, DNA / <i>Taq</i>, polymerase, can bind to DNA or so that, DNA / <i>Taq</i>, polymerase, can replicate or amplify DNA ; <i>how</i> 3 (if) normal primers bind then, normal (genotype) / Hb^AHb^A ; 4 (if) mutant primers bind then, sickle cell anaemia (genotype) / Hb^SHb^S ; 5 (if) both primers bind then, SCT (genotype) / heterozygote / Hb^AHb^S ;	3

Question	Answer	Marks
4(b)(iii)	1 correct genotypes for A and B: A – Hb^A Hb^S accept Hb^N Hb^S and B – Hb^S Hb^S ; 2 correct phenotypes for A and B: A – SCT and B – sickle cell anaemia ;	2
4(c)(i)	any four from: 1 heterozygotes / Hb^A Hb^S / SCT / carriers, have, <u>selective advantage</u> ; 2 malaria is a selection pressure or homozygous dominant / Hb^A Hb^A, die of malaria ; 3 sickle cell anaemia is a selection pressure or homozygous recessive / Hb^S Hb^S, die of sickle cell anaemia ; 4 heterozygotes / Hb^A Hb^S / SCT / carriers, survive / reproduce ; 5 (so) passes on, Hb^S allele / sickle cell allele / mutant allele (to next generation) ; 6 stabilising selection ;	4
4(c)(ii)	any two from: 1 <i>ref. to</i> designer babies / choose sex of baby ; 2 embryos, discarded / destroyed / damaged ; 3 against personal belief / <i>ref. to</i> right to life ;	2